CYBR-2200 Network Analysis
Week 5 Lab Worksheet
Analyzing Attack Patterns & Policy Violations in Network Traffic

Purpose:
This lab focuses on recognizing attack-related network traffic and policy violations. You will analyze packet captures to identify indicators of phishing, unauthorized file sharing, or anonymization services and explain how analysts distinguish malicious behavior from policy violations.

This lab emphasizes interpretation and evidence-based reasoning.

Part A: Open the Provided PCAP Files

1. Download all PCAP files provided in Blackboard for Week 5.
2. Open Wireshark.
3. Open each PCAP file one at a time using File → Open.
4. Confirm packets load successfully for each file.

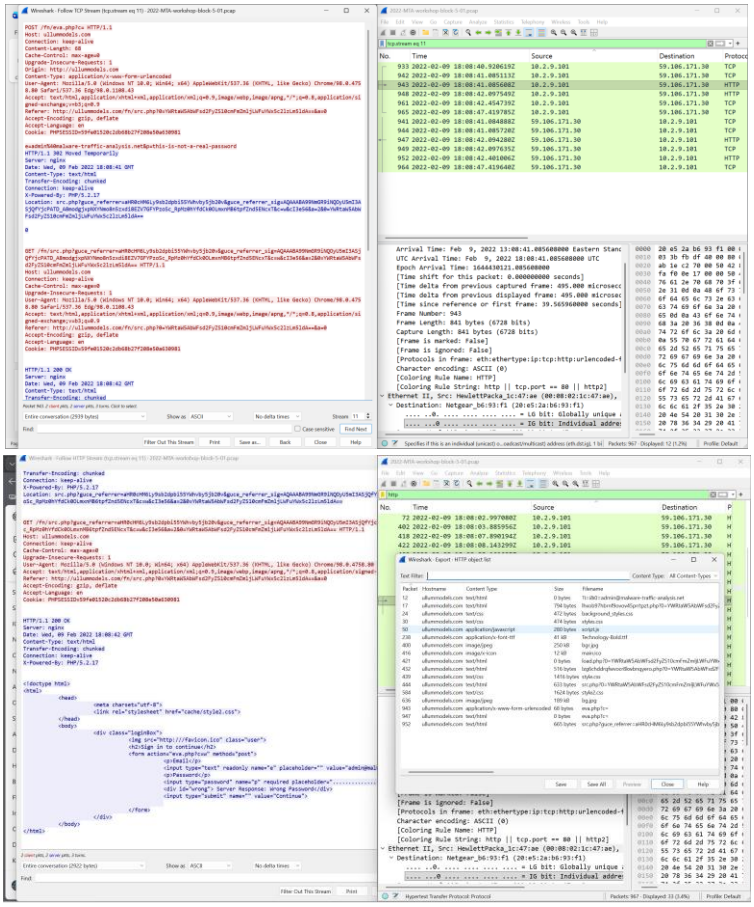
Part B: Identify Suspicious or Policy-Violating Traffic

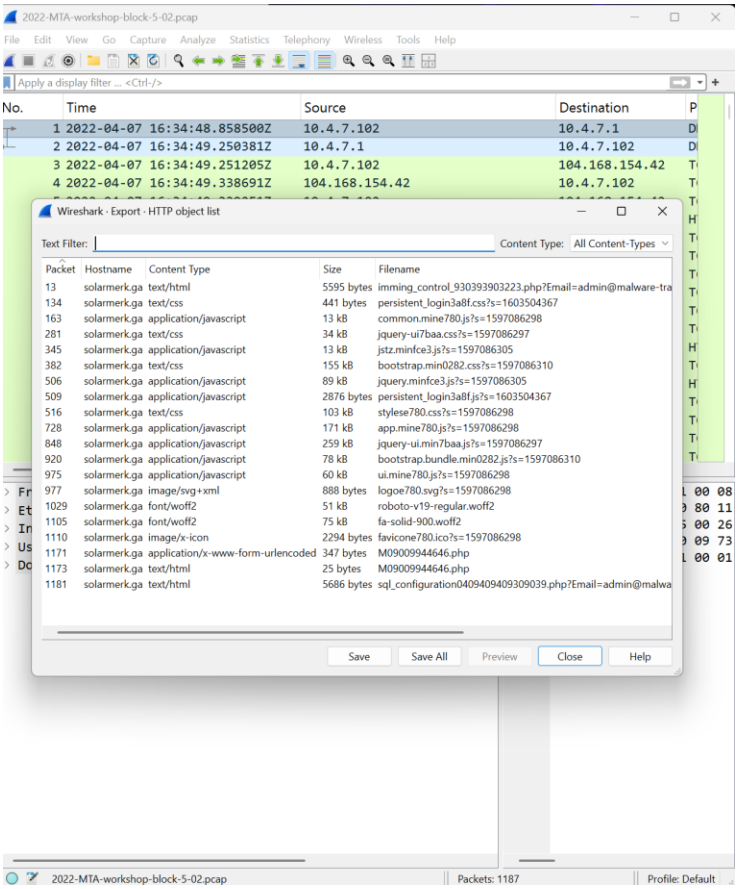
For each PCAP file, identify traffic that may indicate:
• Phishing activity
• Unauthorized file sharing or downloads
• Use of anonymization or tunneling services

Record your findings in the table below.

PCAP File	Traffic Type Observed	Protocol(s)	Source IP	Destination IP / Domain	Why This Is Concerning
2022-MTA-workshop-block-5-01	Phishing through HTTP and credential harvesting	HTTP	Ethernet II, Src: 10.2.9.101 HewlettPacka_1c:47:ae (00:08:02:1c:47:ae)	Dst: 59.106.171.30 Netgear_b6:93:f1 (20:e5:2a:b6:93:f1)	This shows a clear phishing activity with creating an open connection and credential harvesting on packet 943.

2022-MTA-workshop-block-5-02.pcap	Phishing activity was confirmed through a fake login page for solarmerk.ga	HTTP	Internet Protocol Version 4, Src: 10.4.7.102,	Dst: 104.168.154.42	After reviewing the export objects, a Roundcube javascript was a classic phishing attempt.





Part C: Evidence & Interpretation

For EACH example identified above, answer the following:

1. What evidence supports your conclusion?

<u>2022-MTA-workshop-block-5-01-</u> On packet 943, you can see a phishing attempt with a long open connection with credential harvesting. SAGE AI helped analyze and identify the `POST /fn/eva.php?c= HTTP/1.1 Host: ullummodels.com` **along with** `e=admin%40malware-traffic-analysis.net p=this-is-not-a-real-password` **which shows a user submitted an email and password over HTTP unencrypted traffic to a suspicious domain using a fake login form. After searching and decoding the base64, it was determined this was a fake yahoo login account.**

<u>2022-MTA-workshop-block-5-02:</u> This PCAP shows another phishing attempt. Using the same methods as PCAP 1, we were able to see that Packet 1171 was a similar attempt at phishing using solarmerk.ga as the path to a cloud to harvest information. They utilized a Roundcube Javascript to conduct the attack as shown in the exported objects.

2. Does this traffic indicate a confirmed attack or a policy violation? **This traffic confirms an active phishing attack took place by spoofing yahoo.com's login page by looking at the referrer header.**

BLOCK 2 PCAP: This shows a clear phishing attack through credential harvesting hosted by solarmerk.ga and appropriate steps should be taken immediately to isolate the computer from the system until cleaned and restored.

3. What additional information would you need to confirm intent?

BLOCK 1 PCAP: **The additional information you would then need is to look up the DNS logs on the PCAP, browser history to find the URL, and threat intelligence look up to see if the user accessed the site on his own volition or was he redirected from a malicious advertisement or malicious email.**

BLOCK 2 PCAP: **Additional information I would need to look up includes browser history to determine how he got onto the phishing site, either through email logs, DNS logs, and domain lookups on solarmerk.ga. I would also have to use the company's endpoint detection and response software to see if any malware was installed or downloaded. This should help gain a better understanding of how deeply the system was compromised and how to fix it.**

Use complete sentences.

Part D: Analyst Decision-Making

In one short paragraph, explain how an analyst should respond differently to:
• Confirmed malicious activity
• Policy violations that are not necessarily malicious.

There is a major difference between how an analyst should respond to confirmed malicious activity versus policy violations. Confirmed malicious activity means an ongoing attack could be happening as we speak. Therefore, the analyst has a set of protocols to complete when that occurs. They must isolate the affected systems, preserve the evidence, escalate the incident immediately through the proper channels to incident response unit, and begin containment and remediation efforts of the systems and data. You must immediately confirm that no other systems or data were breached, stolen, or altered. Urgency is of upmost importance as the security of the organization could be at risk. On the contrary, policy violations such as unauthorized virtual private networks, file sharing services, or messaging services do pose a risk to the system, but can be handled through proper channels with documentation, notification to the user of unapproved software, and policy enforcement. I would also state that there should be a list of unapproved software that has known vulnerabilities and poses a much more significant threat to the

organization which could be a bridge between the two. If the software in question is on that list, then it can be brought to incident response for immediate removal and scrubbing of the infected computer or hardware.

Submission Instructions

• Save this completed worksheet as a DOCX or PDF file.
• Submit the file in Blackboard under:
Week 5 Lab: Attack Patterns & Policy Violations
• Do NOT upload PCAP files.
• Screenshots are optional but not required.

AI**

SAGE AI was used to help read and understand the PCAPs.